[redacted]

[redacted] CGA-67-6217

PRIORITY: FROZEN SEC

SURGICAL PROCEDURES:
LUNG-LOBECTOMY, RESECTION, REGIONAL LYMPH

CLINICAL HISTORY

PERTINENT CLINICAL HISTORY: LEFT LUNG CA.

GROSS DESCRIPTION

A. The specimen is submitted fresh for frozen section as "9L" and consists of a single red black lymph node measuring 0.6 cm; submitted in toto, 1 (1) labeled "FSP" for frozen section.

FROZEN SECTION DIAGNOSIS: Lymph node: Negative for tumor. [redacted]

B. The specimen is submitted fresh for frozen section as "level 5" and consists of four red black lymph nodes ranging from 0.5 to 0.8 cm; submitted, 4 (1) labeled "FSP" for frozen section.

FROZEN SECTION DIAGNOSIS: Level 5: Negative for tumor. [redacted]

C. The specimen is submitted fresh for frozen section as "level 6" and consists of a single red black lymph node measuring 1.5 cm which is bisected and submitted in toto, 2 (1) labeled "FSP" for frozen section.

FROZEN SECTION DIAGNOSIS: Level 6: Negative for tumor. [redacted]

D. The specimen is submitted fresh for frozen section as "anterior mediastinal mass" and consists of a 2.5 x 1.8 x 1.5 cm roughly spherical lobular mass with tail of yellow adipose tissue exhibiting tan lobulated fleshy tissue and a 0.3 cm area. Upon cut section the external surface is marked with black dye. The specimen is sectioned and submitted in toto,

[redacted]

GROSS DESCRIPTION (Continued) [redacted] TCGA-67-6217

FROZEN SECTION DIAGNOSIS: Anterior mediastinal mass: Favor epithelioid thymoma. Classification pending permanent sections. [redacted]

E. The specimen is submitted fresh but not from frozen section as "left upper lobe" and is a 114 gram of (formalin fixed), 18 x 11 x 2.3 cm lung lobectomy specimen received with a stapled bronchial margin measuring 3 cm in diameter and multiple adjacent stapled vasculature margin measuring 0.5 to 2.3 cm in greatest dimension along with two stapled parenchymal margins measuring 1.7 and 2.6 cm in diameter. The pleural surface is pink to purple red with focal areas of anthracotic pigmentation along with an indurated, firm well circumscribed puckered area. The pleural surface overlying this area is inked blue. The stapled parenchymal margins are removed and the underlying parenchyma is inked black. Sectioning of the firm puckered area reveal a 2.5 x 2 x 1.4 cm well circumscribed gray to white firm mass which grossly appears to abut the overlying blue inked pleural surface measuring 0.4 cm from the closest parenchymal resection margin (black ink) and 0.9 cm from the bronchial margin at the hilum. The remaining parenchyma is tan pink to red spongy and homogenous showing focal areas of emphysematous changes. No other discreet masses or nodularities are grossly identified. At the hilum are multiple gray black hilar lymph nodes measuring 0.5 to 1 cm in greatest dimension, each of which is submitted entirely. Representative sections of the specimen are submitted, multiple (8) as follows:

1 - Vasculature and bronchial margins (submitted en face)
2 - Mass with closest parenchyma underlying stapled resection margin inked black.
3 and 4 - Mass with closest pleural surface.
5 - Mass with adjacent uninvolved parenchyma.
6 - Parenchyma showing emphysematous changes.
7 - Uninvolved parenchyma.
8 - Hilar lymph nodes.

A piece of tumor is collected for TCGA tissue collection project.

F. The specimen is submitted fresh as "posterior interlobular node" and consists of a single gray black elongate lymph node measuring 1 cm which is bisected and submitted in toto, 2 (1).

FINAL DIAGNOSIS

A. LYMPH NODE, 9L, BIOPSY:

- LYMPH NODE, NEGATIVE FOR TUMOR.

[redacted]

[redacted]

FINAL DIAGNOSIS (Continued)

B. LYMPH NODE, LEVEL 5, EXCISION:

- MULTIPLE LYMPH NODES, NEGATIVE FOR TUMOR.

C. LYMPH NODE, LEVEL 6, EXCISION:

- LYMPH NODE, NEGATIVE FOR TUMOR.

D. ANTERIOR MEDIASTINAL MASS, RESECTION:

- THYMOMA (TYPE AB) WITH MICROSCOPIC TRANSCAPSULAR INVASION (2.5 CM), MARGINS INVOLVED. SEE SYNOPTIC REPORT.

E. LUNG, LEFT UPPER LOBE, LOBECTOMY:

- ADENOCARCINOMA (2.5 CM). SEE SYNOPTIC REPORT.
- METASTATIC CARCINOMA (0.3 CM FOCUS) INVOLVING ONE OF FIVE HILAR LYMPH NODE FRAGMENTS.

F. LYMPH NODE, POSTERIOR INTERLOBULAR, EXCISION:

- LYMPH NODE, NEGATIVE FOR TUMOR.

[redacted]

SYNOPTIC REPORT

SYNOPTIC REPORT: THYMOMA

[redacted]

SPECIMEN: THYMUS
PROCEDURE: PARTIAL RESECTION
SPECIMEN INTEGRITY: INTACT
SPECIMEN WEIGHT: APPROXIMATELY 2 GRAMS
TUMOR SIZE: 2.5 CM
HISTOLOGIC TYPE: TYPE AB THYMOMA
TUMOR EXTENSION: NOT IDENTIFIED
MARGINS: INVOLVED. TUMOR ABUTTS ON AND FOCALLY INVOLVES SURGICAL MARGINS.
(SPECIMEN WAS RECEIVED UNORIENTED)
TREATMENT EFFECT: NOT APPLICABLE
LYMPHOVASCULAR INVASION: NOT IDENTIFIED
REGIONAL LYMPH NODES: NONE SAMPLED
IMPLANTS/DISTANT METASTASIS: CANNOT BE ASSESSED
OTHER PERTINENT FINDINGS:

- THE TUMOR IS IMMUNOREACTIVE FOR CYTOKERATIN 5/6 BUT IS NEGATIVE FOR NEUROENDOCRINE MARKERS, SYNAPTOPHYSIN AND CHROMOGRANIN.

PATHOLOGIC STAGE (MODIFIED MASAOKA): IIa (R1)

[redacted]

SYNOPTIC REPORT: LUNG
(CAP/AJCC/ACS - REQUIRED ELEMENTS)

SPECIMEN TYPE: LOBECTOMY
LATERALITY: LEFT
TUMOR SITE: LEFT UPPER LOBE
TUMOR SIZE: 2.5 CM
HISTOLOGIC TYPE: ADENOCARCINOMA
HISTOLOGIC GRADE: GRADE 2 OF 3 (MODERATELY DIFFERENTIATED)
EXTENT OF INVASION: VISCERAL PLEURAL INVASION PRESENT
EXTENTION TO PLEURA/OTHER STRUCTURES: ABSENT

MARGINS:
BRONCHIAL: UNINVOLVED
PARENCHYMAL: UNINVOLVED
VASCULAR: UNINVOLVED
DISTANCE TO PERTINENT MARGIN(S): TUMOR IS 0.4 CM FROM THE PARENCHYMAL RESECTION MARGIN AND 0.9 CM FROM THE BRONCHIAL MARGIN

VENOUS INVASION: ABSENT
ARTERIAL (LARGE VESSEL) INVASION: ABSENT

LYMPH NODE:
f,,,, N1 - # INVOLVED/# EXAMINED: 1/8+
f,,, N2 - # INVOLVED/#EXAMINED: 0/10+
f,,,, N3 - # INVOLVED/# EXAMINED: 0/0

OTHER PERTINENT FINDINGS:
- AN ELASTIC STAIN HIGHLIGHTS VISCERAL PLEURAL INVASION.
- UNINVOLVED LUNG SHOWS MILD EMPHYSEMA.

pTNM STAGE: pT2a pN1

Unless otherwise specified, the performance characteristics of the immunoperoxidase stains performed here have been confirmed by the [redacted]. [redacted]. These tests have not been cleared or approved by the U.S. Food and Drug Administration, although such approval is not necessary.

Patient Information Resources

Thymoma

This section has been reviewed and approved by the [redacted]

Staging

Staging is a way of describing a cancer, such as where it is located, if or where it has spread, and if it is affecting the functions of other organs in the body. Doctors use diagnostic tests to determine the cancer's stage, so staging may not be complete until all the tests are finished. Knowing the stage helps the doctor to decide what kind of treatment is best and can help predict a patient's prognosis (chance of recovery). There are different stage descriptions for different types of cancers.

There is no standard system for staging thymoma; however, the most commonly used system is called the Masaoka system. This system was developed in [redacted] and it classifies thymoma into the following stages:

Stage I: The cancer is limited to the thymus and the capsule that surrounds it.

Stage II: The cancer has spread into fat surrounding the thymus or into the lining of the lung next to the tumor, called the mediastinal pleura.

Stage III: The cancer has spread to other organs that are near the thymus, such as the lung, blood vessels, and the sac around the heart, called the pericardium.

Stage IVA: The cancer has spread more extensively into the lining of the lung or the sac around the heart.

Stage IVB: The cancer has spread to organs further away from the thymus, or has spread through the vessels carrying blood or lymph.

Classification

In addition to its stage, thymoma can be classified into different categories, developed by the [redacted] ([redacted]), based on what the tumor cells look like under a microscope:

Type A thymoma. This is also called spindle cell thymoma or medullary thymoma. The chance of recovery for people with type A thymoma is good, with a survival rate (percentage of people who survive at least after the cancer is detected, excluding those who die from other diseases) near

TypeAB thymoma. Also called mixed thymoma, type AB thymoma is similar to type A thymoma; however, type AB thymoma has lymphocytes in the tumor. The chance of recovery for people with type AB thymoma is also good, with a survival rate of approximately %.

Type B1 thymoma. This is also known as lymphocyte-rich thymoma, lymphocytic thymoma, predominantly cortical thymoma, and organoid thymoma. This type of thymoma has very many lymphocytes in the tumor, but the cells of the thymus appear normal. The chance of recovery for people with type B1 thymoma is also good, with a 20-year relative survival rate (percentage of people who survive at least years after the cancer is detected, excluding those who die from other diseases) of approximately

Type B2 thymoma. Type B2 thymoma also has many lymphocytes, like type B1 thymoma; however, the thymus cells do not appear normal. Type B2 thymoma is also known as cortical thymoma and polygonal cell thymoma. The -year relative survival rate for people with type B2 thymoma is approximately

Type B3 thymoma. Type B3 thymoma is also known as epithelial thymoma, atypical thymoma, squamoid thymoma, and well-differentiated thymic carcinoma. This type of thymoma has few lymphocytes, and the thymus cells look abnormal. The survival rate is approximately

Thymic carcinoma. Thymic carcinoma, previously called type C thymoma, is very rare and more aggressive. Thymic carcinoma cells do not look like normal thymus cells, but like cancers in other organs of the body. This type of tumor is often advanced when diagnosed. It is classified into two categories: low grade, which has a better chance of recovery, and high grade, which is more likely to grow and spread quickly. Low-grade thymic carcinoma includes basaloid, mucoepidermoid, and well-differentiated squamous cell types. High-grade thymic carcinoma includes anaplastic/undifferentiated, clear cell, poorly differentiated squamous cell, sarcomatoid, and small cell/neuroendocrine types. The five-year relative survival rate of people with thymic carcinoma is rate of people with thymic carcinoma is %.

Cancer survival statistics should be interpreted with caution. These estimates are based on data from thousands of cases of this type of cancer in the United States, but the actual risk for a particular individual may differ. It is not possible to tell a person how long he or she will live with thymoma.

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